BB scattering

In a scattering experiment, a projectile flies toward a target and is scattered. The scattering interaction may be a physical collision (e.g., for two billiard balls) or another force (e.g., a Coulomb repulsion or gravitational attraction). Over the course of many such scattering events, the scattering pattern (i.e., the directions of all scattered particles) is recorded in some manner. Finally, the scattering pattern is analyzed in order to give some information about the shape of the target and its interactions with the projectiles. The information provided by a scattering experiment is indirect and incomplete, but it is often the only information we can get about objects we want to study.

Two of the important variables used in scattering discussions are illustrated in Fig. 1. The incoming particles have various values of b, the **impact parameter**, which measures their perpendicular distance from the central axis when they are traveling toward the target. After scattering, the particles travel outward with a **scattering angle** θ, which is the change in angle from their initial velocity. The value of b and the strength of the interaction force jointly determine the scattering angle θ.

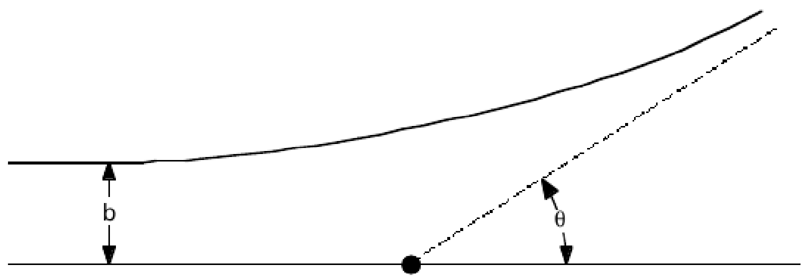

Figure 1: A particle coming in from the left with impact parameter b scatters at angle θ.

In a real scattering experiment, we do not launch particles individually at specific values of b. Instead, we have an incoming "beam" of many particles that each have different (and typically random) values of b spread over the area of the beam, each of which will scatter with a different angle θ. Some, generally smaller, values of b result in large scattering angles; while other, generally larger, values of b lead to small scattering angles. This is illustrated in Fig. 2. There is some area perpendicular to the beam, labeled $\Delta\sigma$ in Fig. 2, that results in scattering into the angular range $\Delta\theta$. This $\Delta\sigma$ is the **cross section** for scattering into $\Delta\theta$. For 3D experiment, $\Delta\sigma$ has units of area because any particles of the beam that pass through that area will have a scattering angle between θ and $\theta + \Delta\theta$

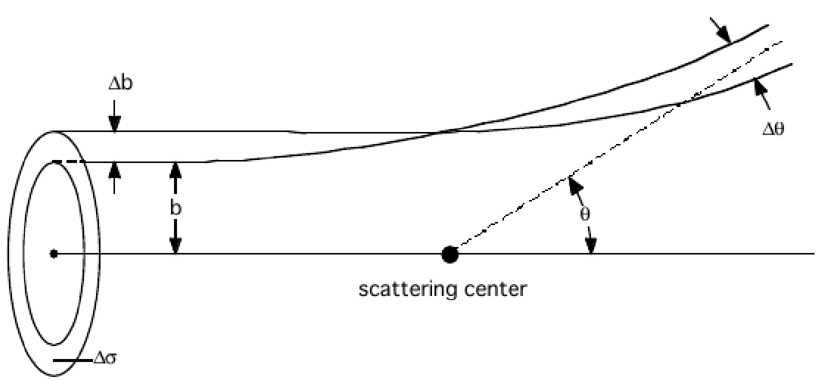

Figure 2: All incoming particles incident on the cross sectional area $\Delta\sigma$ are scattered into the angular range $\Delta\theta$.

The beam intensity is described by the **flux** Φ, which is the number of beam particles per unit area per unit time. That is, if the beam has a cross sectional area of A, the total rate of beam particles per second is $r = \Phi A$. If we want to know the rate at which particles will scatter into our specific range of angles $\Delta\theta$, we should instead multiply the flux by the corresponding cross sectional area $\Delta\sigma$:

$$\Delta r = \Phi \Delta\sigma \tag{1}$$

where Δr describes the number of particles per second that have a scattering angle in the range $\Delta\theta$. If we consider a large $\Delta\theta$ (i.e., a large range of scattering angles) we should expect to capture a large fraction of the total scattered particles. Therefore, we can also write the equation above explicitly in terms of how Δr relates to $\Delta\theta$:

$$\Delta r = \Phi \frac{d\sigma}{d\theta} \Delta\theta \tag{2}$$

where have used the differential relationship $\Delta\sigma = (d\sigma/d\theta)\Delta\theta$, and $d\sigma/d\theta$ is known as the **differential cross section** of the target.

Here we can see that the rate Δr is proportional to $\Delta\theta$, but that this rate also depends on the relationship between σ and θ. If there is only a small area of the target that will cause a specific scattering angle, then we will have fewer particles scattering with that angle. The differential cross section is therefore determined by the physics of the interactions between the incoming particle and the target (i.e., the shape and orientation of the target and the types of forces involved). Of course, if there is more than one target particle (as is usually the case for atomic or nuclear targets), the expressions for Δr must be multiplied by the number of targets, but we will consider a single target here.

In our experiment, we will also simplify things by making all of the motion occur in a horizontal plane (i.e., in 2D rather than 3D). Consequently, the cross section for scattering into a given angular range $\Delta\theta$ is really a cross sectional *length*, not a cross sectional *area*. Likewise, our beam is really a band of particles spread over a width h instead of an area A (see Fig. 3), so the intensity of our beam will be based on the number of particles per unit length rather than per area. Finally, for purposes of this experiment we remove the "per unit time" part of the flux definition since we will be launching our particles one by one and counting them up at the end, rather than having a continuous flow. Based on these simplifications, we will redefine our "flux" as $\Phi = n/h$: the number of particles n that we launch, divided by the length h over which they are spread.

As you think about this particle "beam", it is important to remember that the flux of a beam (whether in 3D or 2D) is determined completely by the properties of the beam and is independent of the particular target that happens to be present. We could imagine putting targets of different shapes and sizes into the same beam and getting different distributions of scattering angles, but the flux Φ of the beam describes the *incoming* distribution of particles before they interact with the target.

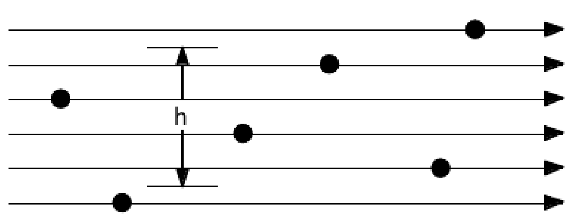

Figure 3: In our experiment the beam of particles is a horizontal band. The beam strength is measured by a "flux" $\Phi = n/h$ equal to the ratio of n, the number of particles, to the length h measured perpendicular to the beam.

The target in this experiment is a circular disk of radius R, which we can use to find the scattering angle θ as a function of b and then the differential cross section $d\sigma/d\theta$ for scattering off of this target. We start with the law of reflection, which states that the incident angle equals the final angle (both measured with respect to the normal). The geometry of this scattering is shown in Fig. 4.

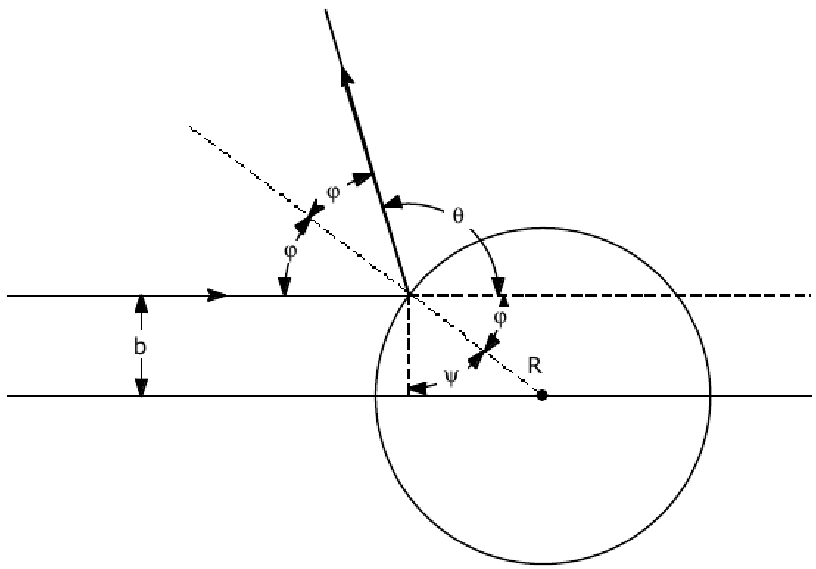

Figure 4: The geometry for a BB of impact parameter b scattering off of a disk of radius R. The scattering angle between the initial and final direction is θ.

From the trigonometry of right triangles, one has $\cos(\psi) = b/R$, so that $\psi = \cos^{-1}(b/r)$. From Fig. 4 we can see that $\phi = \pi/2 - \psi$, so that $\theta = \pi - 2\phi = 2\psi = 2\cos^{-1}(b/R)$ and therefore:

$$b = R\cos(\theta/2) \tag{3}$$

Now that we know the scattering angle at each particular value of b, we can determine the differential cross section by considering how a range of b values (i.e., particles with impact parameters between b and $b + \Delta b$) will scatter into a range of angles between θ and $\theta + \Delta\theta$ as shown in Fig. 5:

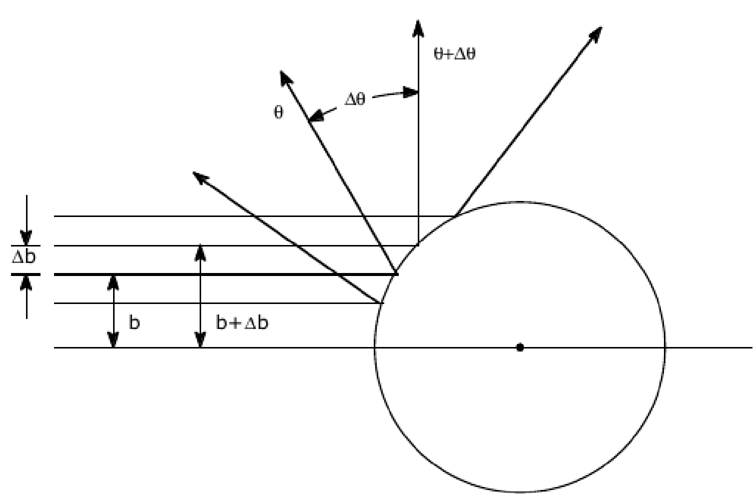

Figure 5: All particles with impact parameter between b and $b + \Delta b$ will be scattered with angle between θ and $\theta + \Delta\theta$. The cross sectional length $\Delta\sigma = (d\sigma/d\theta)\Delta\theta$ is just equal to Δb.

Specifically, the impact parameter that corresponds to scattering at angle $\theta + \Delta\theta$ is:

$$\begin{aligned}
b' &= b + \Delta b && (4)\\
&= b + \frac{db}{d\theta}\Delta\theta && (5)\\
&= R\cos\left(\frac{\theta}{2}\right) + R\left[\frac{d}{d\theta}\cos\left(\frac{\theta}{2}\right)\right]\Delta\theta && (6)\\
&= R\cos\left(\frac{\theta}{2}\right) - \frac{R}{2}\sin\left(\frac{\theta}{2}\right)\Delta\theta && (7)
\end{aligned}$$

Thus the range in b that results in scattering at angles between θ and $\theta + \Delta\theta$ is

$$\Delta b = b' - b = -\frac{R}{2}\sin\left(\frac{\theta}{2}\right)\Delta\theta \tag{8}$$

where the minus sign reflects the fact that a larger b gives a smaller θ. Now the cross sectional length $\Delta\sigma$ for these scattering angle is just range of impact parameters that produces that range of angles:

$$\Delta\sigma = \frac{d\sigma}{d\theta}\Delta\theta = |\Delta b| = \frac{R}{2}\sin\left(\frac{\theta}{2}\right)\Delta\theta \tag{9}$$

Note we have used $|\Delta b$— because the cross section is a positive value (regardless of direction). Consequently, the differential cross section is

$$\frac{d\sigma}{d\theta} = \frac{R}{2}\sin\left(\frac{\theta}{2}\right) \tag{10}$$

The final result for the number of particles N scattered into angular range $\Delta\theta$ at scattering angle θ is

$$N = \Phi\frac{R}{2}\sin\left(\frac{\theta}{2}\right)\Delta\theta \tag{11}$$

Experimental setup

You will shoot a large number of BBs through a hand operated BB gun at a rigid circular target. Data will be recorded as dots on pressure sensitive tape attached to the circular "particle detector" surrounding and concentric with the target. You will analyze this data to determine the radius of the target.

You will use the Welsh BB gun and scattering apparatus with a round plastic target attached, a number of BBs, a meter stick, calipers, masking tape, and a length (about 1.1m) of pressure sensitive tape.

Procedures

1. Turn the position setting screw so that the BBs just miss the target. Shoot a few BBs for practice.

2. Use masking tape to affix the pressure sensitive tape (with the white side showing) to the inside surface of the detector. By left-right symmetry, it is sufficient to study the scattering off of one side of the target instead of both. So, cover only half of the detector with pressure sensitive tape—starting at the gun opening and going in the same direction as the gun's displacement.

3. Shoot five BBs. Turn the position setting screw one-half revolution so that the gun points closer to the target and shoot five more BBs. Continue to shoot five BBs per half revolution until the BBs are bouncing back at the gun. Do not try to keep track of which mark on the tape was produced by which BB in the beam. (This kind of information would not be available in an actual scattering experiment.)

4. Remove the pressure sensitive tape for analysis.

5. Find the "flux" Φ for your beam. Remember that the "flux" is the number of particles fired per unit length (see Fig. 3). It can be found as follows:

   (a) Figure how many particles were fired while the positioning screw was rotated 10 times. Call this number n.

   (b) Measure the length of the range of positions over which these n particles were fired. Call this distance h.

   (c) The "flux" is then $\Phi = n/h$.

6. Locate the $\theta = 0$ point on your tape (see Fig. 6). This is where the BBs just barely miss the target (and so are not scattered). Mark "bins" on your tape starting 1 cm past the $\theta = 0$ point. Make sure each bin has the same width and that you make a note of the bin width you choose, which should be about 5cm. Label them as bin 1, bin 2, etc. starting from the bin closest to $\theta = 0$, and count how many particles N_i scattered into the ith bin for $i = 1, 2, \ldots$ etc.

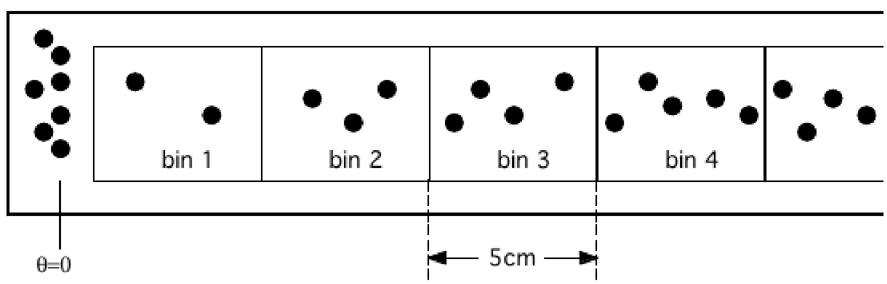

Figure 6: Your data tape will look something like this. The rightmost edge of the accumulation of dots to the left occurs at $\theta = 0$. The angle θ_i of the ith bin is found by measuring the distance between the $\theta = 0$ point and the center of the bin, and then dividing by the detector radius.

7. You now have the data that you will use to analyze the scattering from this experiment! As part of that analysis, you will eventually need to convert the position measurements on your data tape to angles using the relation $\theta = s/R_D$, where s is the arc length along the edge of the detector corresponding to the angle to be measured, and R_D is the radius of the detector. With this in mind, make sure to measure and record the radius of your detector to use in your analysis.